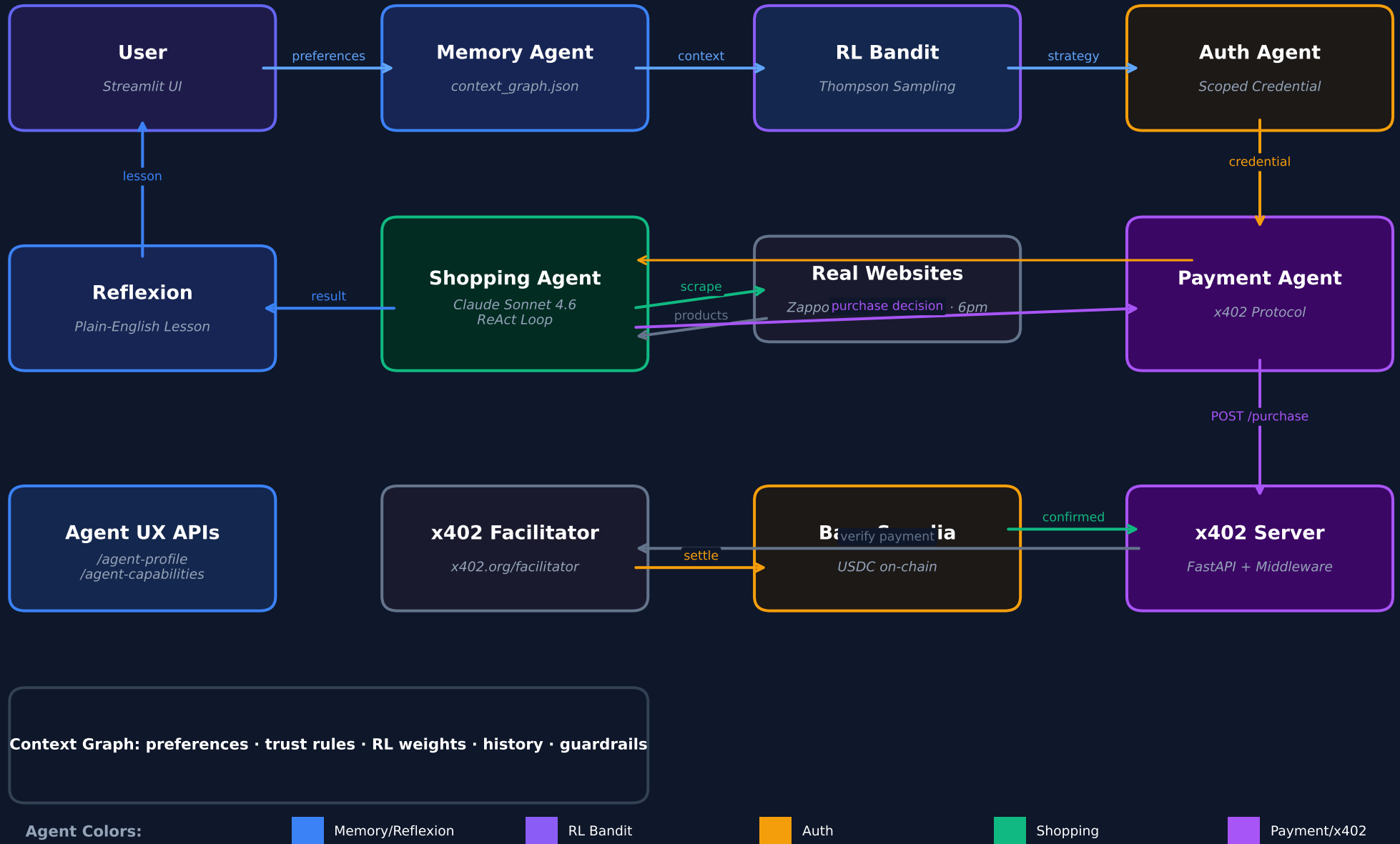


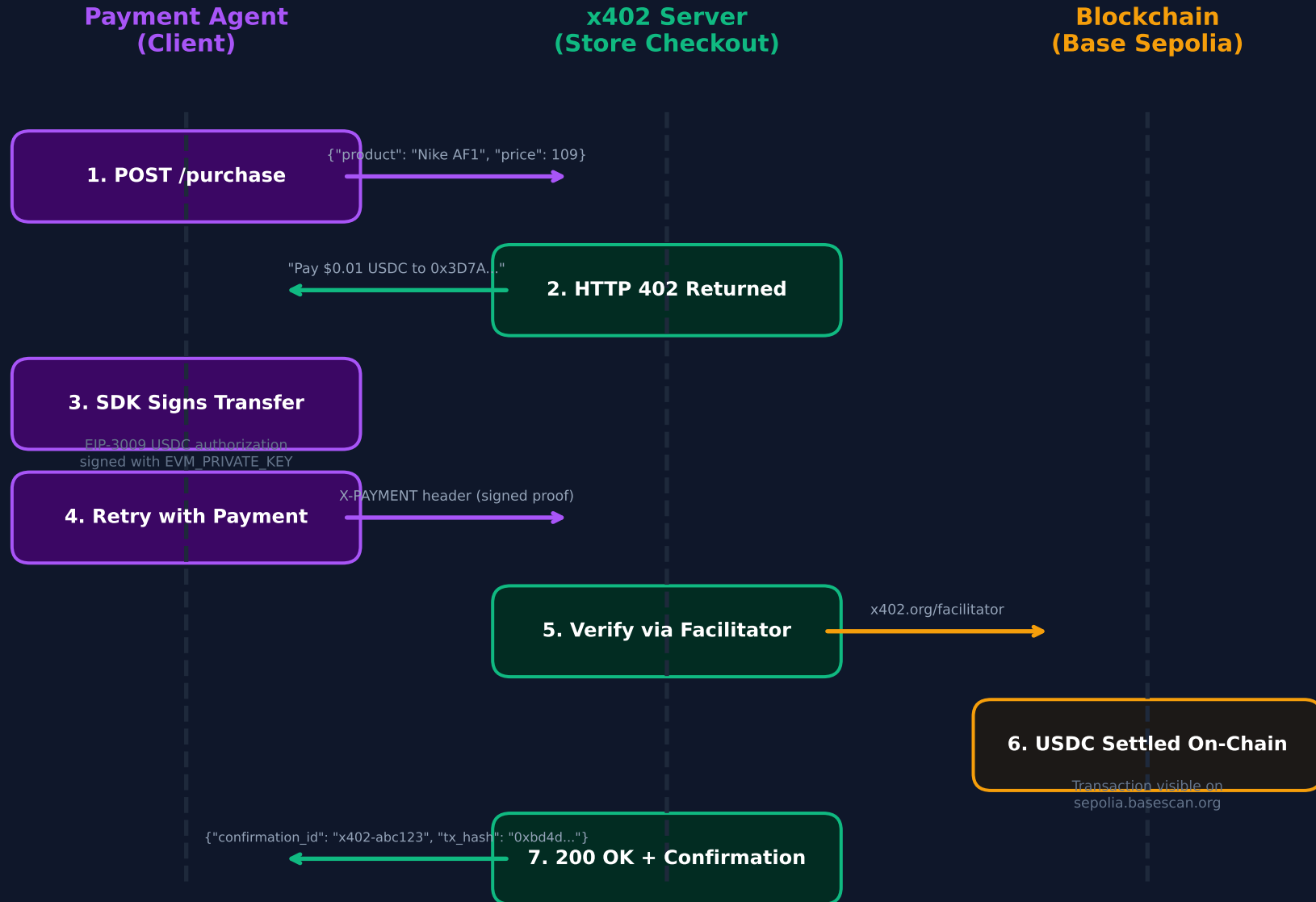
STRIDE — System Architecture

6 Agents · Real Scraping · Thompson Sampling RL · x402 Payments



x402 Payment Flow — How STRIDE Pays

Real USDC on Base Sepolia · Verified by On-Chain Facilitator



STRIDE — Agentic Guardrails

4 Layers of Protection Before Any Purchase

LAYER 1: Preference Guardrails (`shopping_agent.py`)

Color: 'Only buy tiffany blue' · Specific product: 'Nike x Tiffany AF1 only'

Custom instructions: 'Ask before buying anything else' · Excluded brands: 'No Reebok'

LAYER 2: Auth Guardrails (`auth_agent.py`)

Session expiry (1hr) · Site whitelist (6 stores) · Spend cap (\$150)

Budget enforcement (\$120) · Human approval gate (first 3 runs)

LAYER 3: Payment Guardrails (`payment_agent.py`)

Auth clearance required · Double-checks spend cap · Human approval flag

Only fires on final episode (exploration runs don't spend money)

LAYER 4: Protocol Guardrails (`x402_server.py`)

x402 middleware verifies payment signature · Facilitator validates on-chain

Transaction irreversible but transparent · Verifiable on BaseScan

Purchase goes through ONLY if all 4 layers pass